Technique 2: Guesstimate Technique

The "guesstimate" technique is actually more complex than a simple guess at the solution. Rather this technique offers an iterative process where we make an educated guess for b_1 and then solve for a_1, a_2, a_3 similar to how we solve in the two-step process above after estimating b_1. For example, we have found from empirical data that for the most part we have $0.80 \le b_1 \le 1.00$. Thus, we can perform an iterative solution process where we let b_1 vary across these values. For example, allow b_1 to take on each of the values of $b_1 = 0.80, 0.81, \ldots, 1.00$. Then estimate the alpha parameters via non-linear regression analysis. The best fit solution, and thus, model parameter values, can be determined by the best fit non-linear R^2 statistic.

In the case where the temporary impact POV rate parameter $a_4 \neq 1$ we can devise an iterative approach to allow both b_1 and a_4 to vary over their feasible ranges and determine the best fit for all parameters: a_1, a_2, a_3, a_4, b_1. For example,

Let $b_1 = 0.80, 0.81, \ldots, 1.00$

Let $a_4 = 0.00, 0.05, \ldots, 2.00$

Analysts can choose the increment in these iterative loops to be smaller or larger as they feel necessary.

Technique 3: Non-Linear Optimization

Non-linear optimization is very sensitive to the actual convergence technique used, the starting solution, and of course, the level of correlation across the underlying input variables. The solution often has a difficult time distinguishing parameters and we are left with a good model fit but with estimated parameters based on the composite value $k_1 = b_1 \cdot a_1$. While the model may display an accurate fit, it would be difficult to distinguish between the effects of b_1 or a_1 individually. We could end up with a model that fits data well but which is limited in its ability to provide sensitivity and direction. Analysts choosing to solve the parameters of the model via non-linear regression of the full model need to thoroughly understand the repercussions of non-linear regression analysis as well as the sensitivity of the parameters, and potential solution ranges for the parameters. Non-linear regression estimation techniques are the main topic of Chapter 5.

Model Verification

We introduce methods to test and verify results by first forecasting market influences cost and timing risk using estimated parameters. Estimates are compared to actual costs in four different ways.